THE CHALLENGE

FROM TENDER TO FIRST ESTIMATE

Bridge engineering specifications and pricing.

Sprintd Hackathon
9. October 2026

BUILD AN AI-POWERED PRICING ASSISTANT



EQUANS SWITZERLAND

MIBAG PROPERTY MANAGERS · CALIQUA · ENERTRANS · KUMMLER+MATTER | EQUANS DIGITAL · CARBON SHIFT AN EQUANS COMPANY



6’000+

employees in Switzerland

100+

locations

EQUANS SWITZERLAND

WHO WE ARE

BUILDING AND ENERGY SOLUTIONS, FROM ONE SOURCE

We plan, build and maintain everything from smart buildings and tunnels to the power plants of the future.

BUILDINGS

INFRASTRUCTURE

ENERGY

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THE PROBLEM

PRICING A TENDER IS MOSTLY MANUAL

Hundreds of tenders a year. Offers worth millions of Swiss francs.

01



HUNDREDS OF PAGES

Specifications, drawings and bill of quantities to review

02



DATA HUNTING

Line items, products and supplier prices spread across many files

03



MANUAL ESTIMATES

Labor, installation and costs consolidated by hand

The result: experts gather data instead assessing risks and making pricing decisions.

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YOUR MISSION

UPLOAD A TENDER
GET A FIRST ESTIMATE

Automate routine work so estimators can focus on risks and pricing decisions.

01 → 02 → 03 → 04 → 05



INGEST

Tender PDF



EXTRACT

Quantities + units



STRUCTURE

Reviewable table



MATCH

Products, labor and prices



REVIEW

Sources and uncertainty

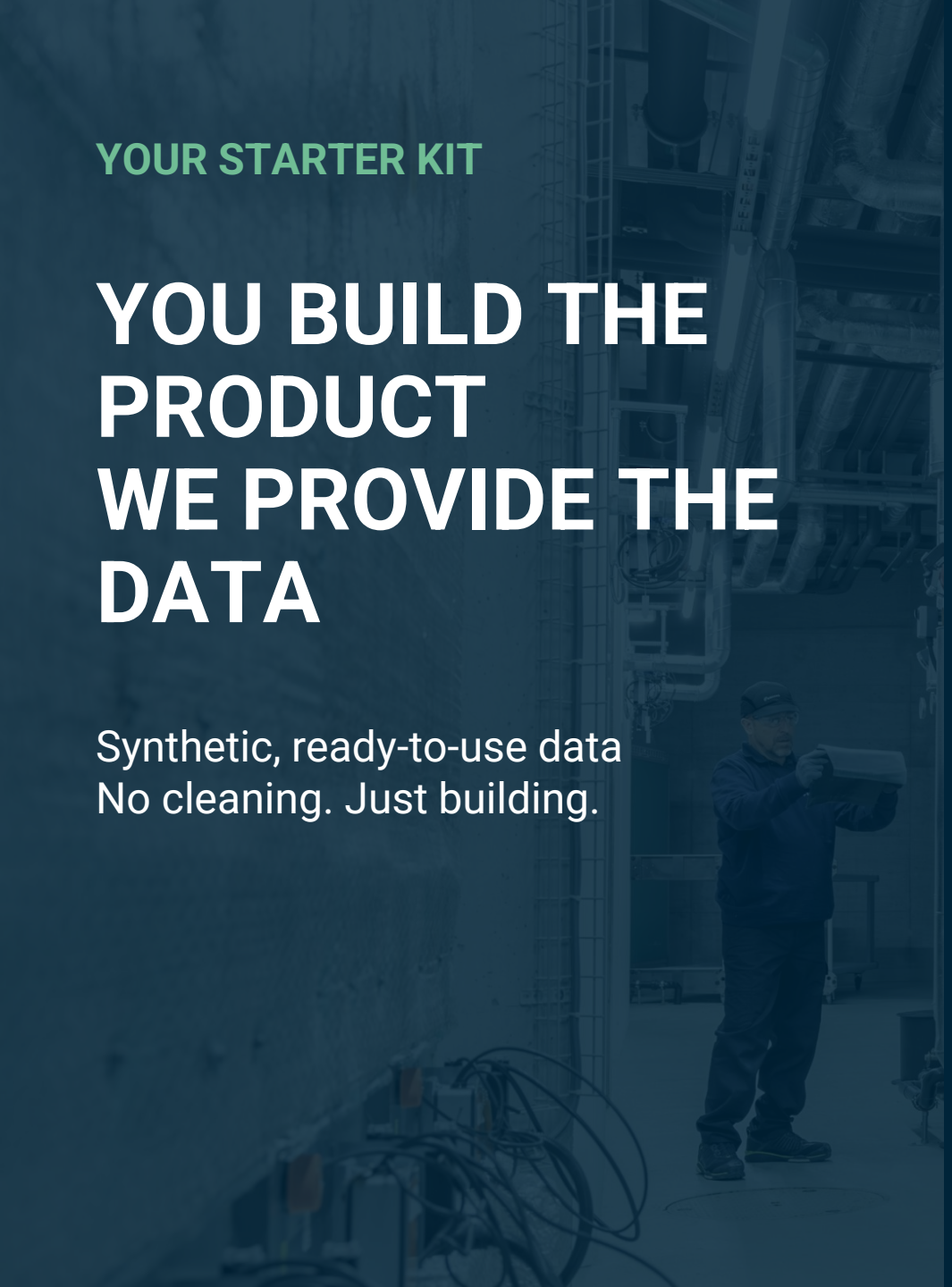
The tool does the groundwork. The expert makes the call.

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YOUR STARTER KIT

YOU BUILD THE PRODUCT WE PROVIDE THE DATA

Synthetic, ready-to-use data
No cleaning. Just building.



01 WHAT YOU READ

TENDER PACKAGE

1 complete tender package

02 WHAT YOU PRICE WITH

PRICING LOGIC

56 product + 22 installation

03 WHAT YOU TEST AGAINST

VERIFIED EXAMPLES

22 test cases

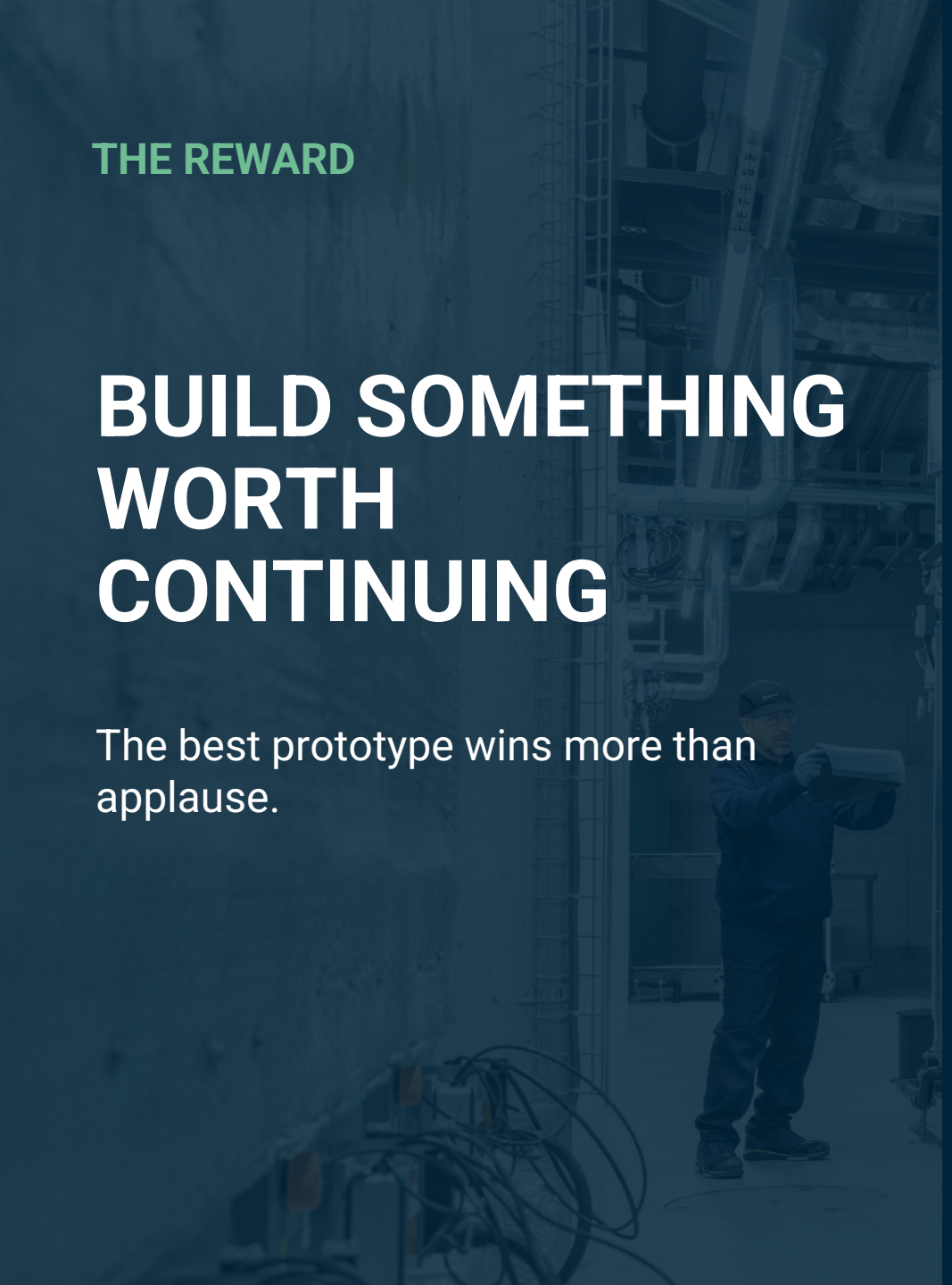
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OUR EXPECTATIONS

BUILD SOMETHING AN ESTIMATOR WOULD ACTUALLY USE

A web app that turns a tender into a transparent first estimate.

01	**USEFULNESS**	Does it solve a real estimator’s problem?
02	**TRANSPARENCY**	Can users trace and review every result?
03	**END-TO-END EXPERIENCE**	Is the journey from tender upload to first estimate clear and functional?
	STRETCH GOAL	Evolve the prototype into a scalable bid optimization platform.



THE REWARD

BUILD SOMETHING WORTH CONTINUING

The best prototype wins more than applause.

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01

CHF 150

Galaxus Voucher

For each member of the winning team

02

Keep building

Continue developing the project with Equans Switzerland

Through an **INTERNSHIP** or **COLLABORATION**

SPRINTD HACKATHON | OCTOBER 2026

From Tender to First Price

Build an AI-powered pricing assistant

Turn a complex public tender into a transparent, reviewable and commercially useful first price.



Schulhaus Oberglatt | BKP 244 Lüftungsanlagen

EQUANS SWITZERLAND PRICING ASSISTANT

Your mission
Design the future of tender pricing at Equans Switzerland and demonstrate how estimators could create transparent and competitive offers in a fraction of the time.

1 Why This Challenge Matters

Today, an estimator must...

- Navigate through multiple tender, pricing and calculation documents
- Manually collect and consolidate relevant information
- Apply experience and expertise to develop a competitive offer

With a pricing tool, an estimator can...

- Access all relevant information in one place
- Start from a structured and pre-filled proposal
- Focus on pricing decisions rather than data consolidation

2 Product Vision



End-to-end flow from tender upload to a transparent pricing and margin overview

3 What You Should Build

- Modern and intuitive user interface
- End-to-end pricing workflow
- Effective support for estimator decision making
- Automatic pre-filling using mock data
- Clickable prototype or working PoC

4 Available Data & Inputs

We provide 2-3 mock project packages containing:

- Bill of Quantities (Leistungsverzeichnis)
- Material Price Lists
- Calculation Sheets

5 What Matters Most

Intuitive user experience
A workflow that estimators can understand and use immediately

Clear end-to-end journey
From tender upload to priced offer in a logical and transparent way

Compelling product vision
A solution that demonstrates the future of tender pricing at Equans Switzerland

6 What Not To Focus On

- Perfect Pricing Accuracy
- Complex AI or ML models
- Production-ready architecture

Prioritise the estimator experience over technical sophistication

1 | OPPORTUNITY

HVAC tender pricing is still manual, fragmented and expert-dependent

The Oberglatt school tender makes the pain visible and the opportunity tangible.

Today: fragmented work

- Navigate long tender documents
- Find relevant positions and requirements
- Search products and supplier prices
- Estimate installation effort
- Consolidate multiple cost categories
- Apply expert judgement to reach a competitive offer

Tomorrow: pricing decisions

- All relevant information in one place
- Structured and pre-filled proposal
- Transparent calculations and assumptions
- Human review where confidence is low
- More time for pricing strategy
- A workflow scalable to other tenders and contractors



THE OPPORTUNITY | Transform an old industry problem into a reusable digital product for HVAC tendering.

2 | CHALLENGE

Process the full tender and create a transparent first price

Cover the full LV and clearly separate calculated, assumed, uncertain and out-of-scope items.

1	**Ingest**	Read and structure the full tender PDF
2	**Extract**	Identify systems, positions, quantities, units and attributes
3	**Match**	Connect positions to catalogue items or supplier packages
4	**Estimate**	Calculate installation hours and labour cost
5	**Price**	Build material, labour, project costs and indicative selling price
6	**Review**	Flag uncertainty and provide traceable output

FULL-LV SUCCESS CRITERION

Every relevant position gets a clear outcome

- ✓ Calculated
- ✓ Supplier quote required
- ✓ Assumption applied
- ✓ Review required
- ✓ Insufficient information
- ✓ Out of scope / non-priceable

DATA PACKAGE

Provided Files

BKP_244_LV_Lueftung_260804.pdf

Tender document: item descriptions, quantities, units and source pages.

catalog.csv

Product catalogue: product details, net costs and installation rule codes.

productivity.csv

Installation rules: base hours, labour roles, hourly rates and factor keys.

project_parameters.json

Project factors, allowances, project costs and commercial parameters.

example_lv_positions.csv

Verified example items for checking extraction and product matching.

expected_output_example.json

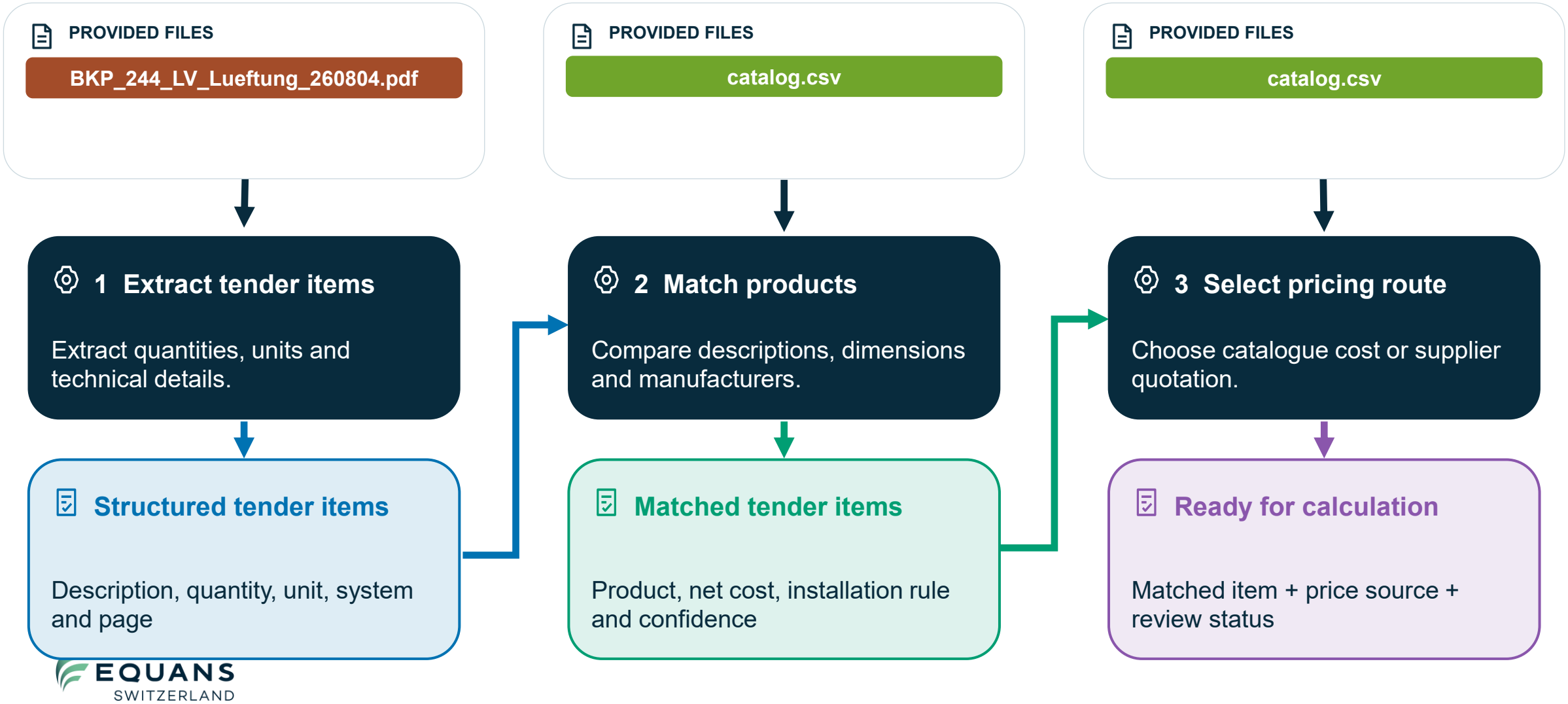
Reference results for the example subset: hours, costs and review status.

README.md

Challenge context and guidance for using the supplied data package.

WORKFLOW 1/3

From Tender to Matched Items

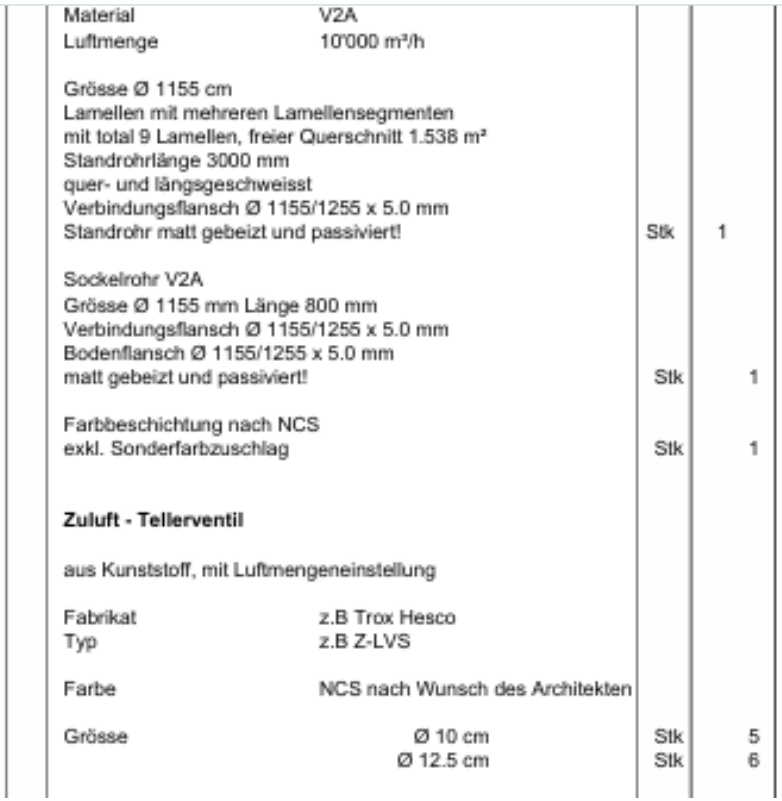


WORKFLOW 1/3

From Tender to Matched Items

BKP_244_LV_Lueftung_260804.pdf

Tender document: item descriptions, quantities, units and source pages.



Material	V2A		
Luftmenge	10'000 m³/h		
Grösse Ø 1155 cm Lamellen mit mehreren Lamellensegmenten mit total 9 Lamellen, freier Querschnitt 1.538 m² Standrohrlänge 3000 mm quer- und längsgeschweisst Verbindungsflansch Ø 1155/1255 x 5.0 mm Standrohr matt gebeizt und passiviert!		Stk	1
Sockelrohr V2A Grösse Ø 1155 mm Länge 800 mm Verbindungsflansch Ø 1155/1255 x 5.0 mm Bodenflansch Ø 1155/1255 x 5.0 mm matt gebeizt und passiviert!		Stk	1
Farbbeschichtung nach NCS exkl. Sonderfarbzuschlag		Stk	1
Zuluft - Tellerventil			
aus Kunststoff, mit Luftmengeneinstellung			
Fabrikat	z.B Trox Hesco		
Typ	z.B Z-LVS		
Farbe	NCS nach Wunsch des Architekten		
Grösse	Ø 10 cm	Stk	5
	Ø 12.5 cm	Stk	6

	lv_id	system	source_page	category	description	quantity	unit
1	LV0001	244.01 LA01 Nebenraeume UG	78	Air terminal	Supply air valve, plastic, airflow adjustable, DN100	5	pc
2	LV0002	244.01 LA01 Nebenraeume UG	78	Air terminal	Supply air valve, plastic, airflow adjustable, DN125	6	pc
3	LV0003	244.01 LA01 Nebenraeume UG	79	Air terminal	Exhaust air valve, plastic, airflow adjustable, DN100	5	pc
4	LV0004	244.01 LA01 Nebenraeume UG	79	Fire protection	Fire damper EI90-S with actuator and interface, DN100	4	pc
5	LV0005	244.01 LA01 Nebenraeume UG	80	Damper	Manual balancing damper, galvanized, DN125	6	pc
6	LV0006	244.01 LA01 Nebenraeume UG	80	Silencer	Rectangular silencer 400x300x1250 mm	4	pc
7	LV0007	244.02 LA02 Essenraeume EG	106	Flow controller	Variable volume flow controller 500x300 mm with actuator and Modbus	4	pc
8	LV0008	244.02 LA02 Essenraeume EG	107	Flow controller	Constant volume flow controller DN250	6	pc
9	LV0009	244.02 LA02 Essenraeume EG	107	Silencer	Compact silencer DN100, length 900 mm	6	pc
10	LV0010	244.02 LA02 Essenraeume EG	107	Silencer	Rectangular duct silencer 600x600x1750 mm	4	pc
11	LV0011	244.04 LA04 Schulzimmer OG	163	Silencer	Rectangular silencer 540x400x1250 mm	6	pc
12	LV0012	244.04 LA04 Schulzimmer OG	163	Flow controller	Variable volume flow controller 300x200 mm	6	pc
13	LV0013	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN125	10	pc
14	LV0014	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN160	6	pc
15	LV0015	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN200	12	pc
16	LV0016	244.04 LA04 Schulzimmer OG	169	Insulation	External thermal duct insulation, mineral wool 30 mm	52	m2
17	LV0017	244.04 LA04 Schulzimmer OG	169	Insulation	External thermal duct insulation, mineral wool 60 mm	12	m2
18	LV0018	244.04 LA04 Schulzimmer OG	169	Insulation	Weatherproof external insulation 100 mm with stainless steel cladding	440	m2
19	LV0019	244.04 LA04 Schulzimmer OG	170	Fire protection	EI30 penetration sealing for rectangular duct 300x200 mm	4	pc
20	LV0020	244.04 LA04 Schulzimmer OG	170	Fire protection	EI30 penetration sealing for circular duct DN160	8	pc
21	LV0021	244.04 LA04 Schulzimmer OG	171	Support	Connection angles for wall or ceiling mounting	8	pc
22	LV0022	244.04 LA04 Schulzimmer OG	171	Fire protection	Inspection cover for fire insulation, duct 400x200 mm	8	pc
23							

Structured tender items

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WORKFLOW 1/3

From Tender to Matched Items

	lv_id	system	source_page	category	description	quantity	unit
1	LV0001	244.01 LA01 Nebenraeume UG	78	Air terminal	Supply air valve, plastic, airflow adjustable, DN100	5	pc
2	LV0002	244.01 LA01 Nebenraeume UG	78	Air terminal	Supply air valve, plastic, airflow adjustable, DN125	6	pc
3	LV0003	244.01 LA01 Nebenraeume UG	79	Air terminal	Exhaust air valve, plastic, airflow adjustable, DN100	5	pc
4	LV0004	244.01 LA01 Nebenraeume UG	79	Fire protection	Fire damper EI90-S with actuator and interface, DN100	4	pc
5	LV0005	244.01 LA01 Nebenraeume UG	80	Damper	Manual balancing damper, galvanized, DN125	6	pc
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12	LV0012	244.04 LA04 Schulzimmer OG	163	Flow controller	Variable volume flow controller 300x200 mm	6	pc
13	LV0013	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN125	10	pc
14	LV0014	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN160	6	pc
15	LV0015	244.04 LA04 Schulzimmer OG	164	Flow controller	Variable volume flow controller DN200	12	pc
16	LV0016	244.04 LA04 Schulzimmer OG	169	Insulation	External thermal duct insulation, mineral wool 30 mm	52	m2
17	LV0017	244.04 LA04 Schulzimmer OG	169	Insulation	External thermal duct insulation, mineral wool 60 mm	12	m2
18	LV0018	244.04 LA04 Schulzimmer OG	169	Insulation	Weatherproof external insulation 100 mm with stainless steel cladding	440	m2
19	LV0019	244.04 LA04 Schulzimmer OG	170	Fire protection	EI30 penetration sealing for rectangular duct 300x200 mm	4	pc
20	LV0020	244.04 LA04 Schulzimmer OG	170	Fire protection	EI30 penetration sealing for circular duct DN160	8	pc
21	LV0021	244.04 LA04 Schulzimmer OG	171	Support	Connection angles for wall or ceiling mounting	8	pc
22	LV0022	244.04 LA04 Schulzimmer OG	171	Fire protection	Inspection cover for fire insulation, duct 400x200 mm	8	pc
23							

Structured tender items

catalog.csv

Product catalogue: product details, net costs and installation rule codes.

5	P-DAM-001	Damper	Manual damper	DemoDuct	MD-125	Manual balancing damper DN125	pc
6	P-SIL-001	Silencer	Rectangular	DemoAcoustic	RS-400-300-1250	Rectangular silencer 400x300x1250 mm	pc
7	P-VAV-001	Flow controller	VAV rectangular	DemoControl	VAV-500-300	VAV controller 500x300 mm with actuator	pc
8	P-CAV-001	Flow controller	CAV circular	DemoControl	CAV-250	Constant volume controller DN250	pc
9	P-SIL-002	Silencer	Compact	DemoAcoustic	CS-100-900	Compact silencer DN100 length 900 mm	pc
10	P-SIL-003	Silencer	Rectangular	DemoAcoustic	RS-600-600-1750	Rectangular silencer 600x600x1750 mm	pc
11	P-SIL-004	Silencer	Rectangular	DemoAcoustic	RS-540-400-1250	Rectangular silencer 540x400x1250 mm	pc
12	P-VAV-002	Flow controller	VAV rectangular	DemoControl	VAV-300-200	VAV controller 300x200 mm	pc
13	P-VAV-003	Flow controller	VAV circular	DemoControl	VAV-125	Variable volume controller DN125	pc
14	P-VAV-004	Flow controller	VAV circular	DemoControl	VAV-160	Variable volume controller DN160	pc
15	P-VAV-005	Flow controller	VAV circular	DemoControl	VAV-200	Variable volume controller DN200	pc
16	P-INS-001	Insulation	Thermal 30 mm	DemoInsulation	MW-30	30 mm mineral wool duct insulation	m2
17	P-INS-002	Insulation	Thermal 60 mm	DemoInsulation	MW-60	60 mm mineral wool duct insulation	m2
18	P-INS-003	Insulation	Weatherproof 100 mm	DemoInsulation	MW-100-SS	100 mm external insulation with cladding	m2

Matched tender items

Product, net cost, installation rule and confidence

Ready for calculation

Matched item + price source + review status

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WORKFLOW 2/3

From Matched Items to Direct Costs

PROVIDED INPUTS

- Step 3 output: quantity + net unit cost
- project_parameters.json
- allowances: material waste

4 Calculate material cost

Quantity x net cost x (1 + waste percentage)

Material cost per item

Calculated material cost

PROVIDED INPUTS

- Step 3 output: quantity + installation rule
- productivity.csv
- project_parameters.json
- productivity_factors

5 Calculate labour cost

Rule hours x quantity x factors x hourly rate

Labour cost per item

Hours, labour role, hourly rate and cost

PROVIDED INPUTS

- Step 4: material cost per item

6 Combine item costs

Add material and labour; retain item and page references.

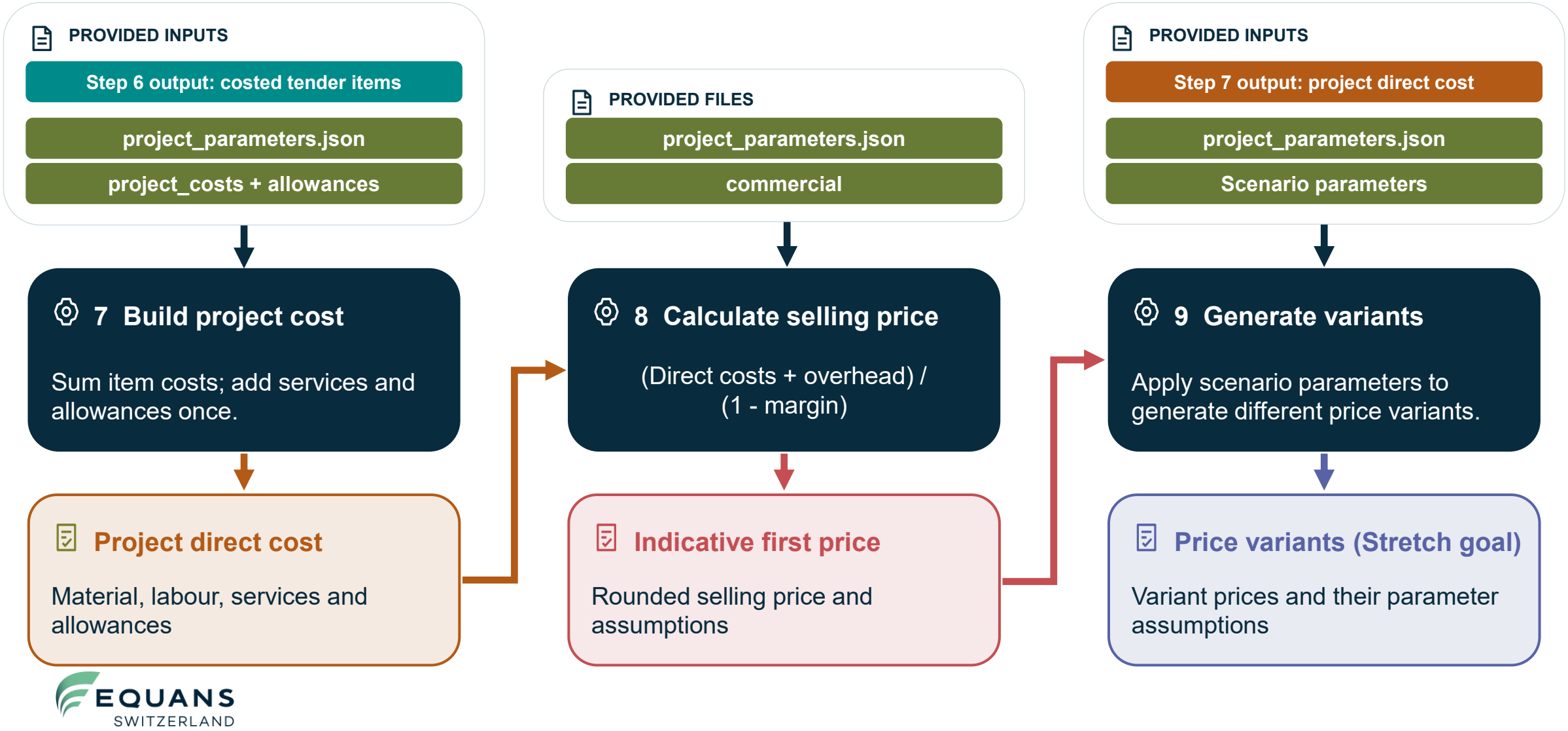
Costed tender items

Material, labour, direct cost and review status

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WORKFLOW 3/3

From Direct Costs to Price Variants



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4 | CALCULATION LOGIC

Every number should be traceable from LV position to selling price

Use the provided catalogue, productivity rules and project parameters.

1 **Material cost**

Quantity × net unit cost × material allowance

2 **Installation hours**

Quantity × base hours × project factors

3 **Labour cost**

Adjusted hours × hourly rate

4 **Direct position cost**

Material + labour

PROJECT-LEVEL BUILD-UP

Material net cost

+ Supplier packages

+ Installation labour

+ Engineering / project management

+ Transport & equipment

+ Special costs

= Direct cost

+ Risk allowance / – Opportunities

+ Overhead

= Cost baseline

÷ (1 – target margin)

INDICATIVE SELLING PRICE

Guardrail: visible assumptions and review flags are better than false precision.

5 | SUCCESS

The strongest prototype combines automation, transparency and human control

Prioritise the estimator experience over technical sophistication.

MINIMUM VIABLE

- ✓ Process the full LV
- ✓ Extract positions and quantities
- ✓ Match some products
- ✓ Calculate material and labour
- ✓ Show source-page references
- ✓ Flag uncertainty

TARGET

- ✓ Process most relevant positions
- ✓ Catalogue + supplier packages
- ✓ Estimate installation hours
- ✓ Aggregate by system/category
- ✓ Human review interface
- ✓ Indicative project price

STRETCH

- ✓ **Bid optimization** + pricing range
- ✓ Alternative product suggestions
- ✓ Supplier RFQ generation
- ✓ Design consistency checks
- ✓ Scenario comparison
- ✓ Export priced offer

JUDGING LENS

Usefulness
Could an estimator work with it?

Traceability
Can every result be understood?

Execution
Does the journey work end-to-end?

UX
Is it fast and intuitive?

Scalability
Could it extend to other tenders?

6 | STRETCH GOAL

Optimize the bid, not only the price

Make the trade-off between bid price, margin and estimated win probability visible.

BID OPTIMIZATION

1. Generate a recommended pricing range
2. Create Aggressive, Target and Conservative scenarios
3. Attach margin and estimated win probability to each price
4. Consider competition, pricing power and strategic appetite
5. Make assumptions visible and editable

	Bid price	Margin	Estimated win probability
AGGRESSIVE	CHF 950k	2.5%	75%
TARGET	CHF 1.00m	4.0%	55%
CONSERVATIVE	CHF 1.08m	6.5%	35%

PRICE ↔ MARGIN ↔ ESTIMATED WIN PROBABILITY

The objective is not one perfect price. It is a better bid decision.

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